

SERCAN OZKAN

Istanbul, Turkey | +90 546 536 31 21 | ozkansercan55@gmail.com
github.com/SercanOzkan55 | cvanalyzer.dev

PROFESSIONAL SUMMARY

Computer Engineering student with hands-on experience in backend development, AI/LLM-powered applications, and real-time systems. Skilled in Python, Java, C#, React, FastAPI, JavaScript, and TypeScript. Built end-to-end products including a multi-runtime CV analysis SaaS platform and a VR educational assistant powered by a RAG (Retrieval-Augmented Generation) pipeline. Experienced with LLM integrations (OpenAI, Gemini, Claude, DeepSeek) and with PostgreSQL, Redis, Celery, and Docker. Seeking a Software Engineering, Backend Engineering, or AI Engineering internship / junior developer role.

EXPERIENCE

Intern Software Engineer | Dasal Havacilik

Istanbul, Turkey

July 2025 – August 2025

- Worked on low-latency drone video transmission: researched and compared streaming protocols (RTSP, RTP/RTCP, TCP vs UDP) and applied them in live streaming tests.
- Set up an RTSP server and streamed live camera feeds over Wi-Fi using both TCP and UDP transports; tested pipelines with FFmpeg, GStreamer, and VLC.
- Measured end-to-end stream latency, analyzed packet loss and network stability, and tuned bitrate/resolution parameters to improve video quality and delay.
- Configured IP/port and network settings on ground control station software and debugged camera connection issues; contributed to C++ stream/GUI integration work.
- Prepared a technical handover document covering streaming tools, connection procedures, latency measurement methods, and troubleshooting for future interns.

PROJECTS

CV Analyzer Platform — cvanalyzer.dev (live)

Python, FastAPI, React, PostgreSQL, Redis

- Built an AI-powered resume analysis platform (hybrid SaaS + local desktop) for candidates and recruiters, spanning four runtimes: FastAPI backend, React + Vite web portal, Expo React Native mobile app, and a PySide6/QML offline batch worker.
- Implemented ATS scoring with per-section breakdowns and a deterministic-first parsing pipeline that falls back to LLMs (OpenAI, Gemini, Claude, DeepSeek) only when needed.
- Developed recruiter workflows: job and batch management, candidate ranking with combined semantic + keyword matching, and embeddings-based similar-candidate search.
- Designed asynchronous large-batch CV processing with Celery and Redis, JWT authentication, plan-based quotas, Alembic migrations, and Dockerized deployment.
- Added a template-based CV builder (DOCX/PDF/HTML export), analysis history, sharing, cover-letter and interview-prep tools, and an analytics dashboard.

VR AI Teacher

Unity (C#), FastAPI, RAG, ChromaDB, OpenAI

- Developed a VR educational assistant in which a 3D teacher avatar answers spoken student questions grounded in course PDF documents.
- Built a RAG pipeline: PDF ingestion and chunking with PyMuPDF, vector embeddings stored in ChromaDB, similarity retrieval, and grounded answer generation via FastAPI.
- Implemented the full voice loop in Unity: microphone speech-to-text, backend query, and text-to-speech playback with viseme-based lip-sync and avatar animation.
- Created an automated benchmark system that scores answers on faithfulness, relevance, correctness, and latency; containerized the backend with Docker (Railway deployment).

FoodApp

PHP, MySQL, HTML, CSS

- Developed a role-based food ordering web application with separate admin, seller, and customer panels.
- Implemented authentication, menu management, shopping cart, checkout, order tracking, favorites, and rating features backed by a MySQL database.

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, JavaScript, TypeScript, PHP, SQL

Frontend: React, Vite, HTML, CSS

Backend:	FastAPI, REST APIs, SQLAlchemy, Celery, JWT Authentication
Databases:	PostgreSQL, MySQL, Redis, SQLite, ChromaDB (vector database)
AI / LLM:	OpenAI (ChatGPT), Gemini, Claude, DeepSeek APIs; RAG pipelines, embeddings, prompt engineering
Tools & Platforms:	Git, Docker, Unity, FFmpeg, GStreamer, AWS S3, Railway, VS Code, IntelliJ IDEA
Concepts:	Data Structures, Algorithms, OOP, Client-Server Architecture, Real-Time Systems, Scalable Systems
Protocols:	TCP, RTP, RTSP, HTTP/REST

CERTIFICATIONS

- **Career Essentials in Generative AI** | Microsoft & LinkedIn | July 2026
- **AWS Cloud Technical Essentials** | edX | April 2026
- **Machine Learning and AI with Python** | edX | March 2026
- **Introduction to Cloud Science** | IBM | March 2026

EDUCATION

Istanbul Health and Technology University	2022 – Present
<i>B.Sc. in Computer Engineering</i>	<i>GPA: 3.82 / 4.00</i>
B.Sc. in Industrial Engineering (Transferred)	<i>GPA: 3.41 / 4.00</i>

LANGUAGES

English: B1+ Speaking | B2 Writing | B2/C1 Technical Reading